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Project Title:	Controlled Assembly of Amphiphilic Janus Particles in Polymer Matrix for Novel 3D Printing Applications In Space		
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Comments:			
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Task Description:	This proposal focuses on developing and analyzing soft materials composed of amphiphilic Janus particles within polymer matrices. It details a research plan for synthesizing Janus particles of both colloidal and larger sizes using modified Stöber methods and stopflow lithography. These particles are characterized by having two distinct sides with different affinities – hydrophilic and hydrophobic – that enable unique aggregation behaviors. The project aims to investigate the self-assembly of these particles in polymer matrices under various conditions, such as temperature, pH, and external fields, to customize material properties for space applications. Additionally, the project includes both experimental and computational studies to optimize the assembly of Janus particles in the presence of electric and magnetic fields, with the goal of enhancing the mechanical properties of 3D-printed materials. This involves identifying optimal conditions for particle assembly during the 3D printing process		

	and creating composite materials for mechanical testing. Brownian Dynamics simulations will offer insights into the behavior of these particles within polymer matrices, informing the experimental approach. The anticipated materials are designed to address existing limitations in 3D printing, including speed, resolution, and material quality, by providing customizable properties such as strength, thermal stability, and radiation resistance, critical for space environments. This initiative contributes to advancing material science and 3D printing technology, aligning with NASA's objectives for space exploration by improving the durability and adaptability of space missions with innovative materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research is directly relevant to Strategy 1.1 of Priority 1 (Exploration and Scientific Discovery) under the NASA Science Mission Directorate (SMD). The Biological and Physical Sciences (BPS) pioneers scientific discovery and enables space exploration by using space environments to conduct investigations not possible on Earth. The potential advancements in material science and 3D printing technology from this project could enhance NASA's capacity for deep-space exploration and in-situ resource utilization. Advancing materials and manufacturing technologies are pivotal for on-demand construction of essential infrastructure and components during space missions. The improved 3D printing capabilities envisioned through this research could forward the development of in-situ manufacturing solutions, facilitating the long-term human habitation of other celestial bodies, and thus supporting the broader objectives of SMD. Furthermore, alignment with the SMD's priorities is demonstrated by the study's capacity to explore new amphiphilic colloidal systems and create materials with specific characteristics that could be beneficial on Earth as well as in space. By combining computational insights with actual observations, experimental design can be made more effective and efficient while also promoting a better understanding of the fundamental ideas of colloidal systems. The study's investigations are in alignment with SMD's priorities: Exploration and Scientific Discovery, Innovation, Interconnectivity and Partnerships, and Inspiration.
Task Progress:	During this reporting period, the project made substantial progress toward developing a new materials approach for strengthening polymer components produced by 3D printing for space-related applications. The central goal of the work is to incorporate amphiphilic Janus particles into polymer resins and use externally applied electric and magnetic fields to organize those particles before curing. Because Janus particles have two chemically distinct sides, they can respond directionally to applied fields and form controlled microstructures. These organized particle networks are expected to provide a pathway for improving the mechanical performance of printed polymer parts, which is important for future manufacturing, repair, and materials-processing capabilities in space. The first year of the project focused on establishing the experimental and theoretical foundation needed for later testing in polymer resin systems. On the experimental side, the team's objectives were to obtain the necessary equipment and materials, develop a reliable method for fabricating silica/platinum Janus particles, and begin mapping how those particles assemble under alternating-current electric fields. The project successfully completed these first-year goals. Materials and equipment were obtained, including silica particles, UV-curable resin, molding materials, a waveform generator, and a high-voltage waveform amplifier. Together, these instruments enabled the team to apply controlled AC electric fields over a range of frequencies and field strengths suitable for studying particle alignment and assembly. A major accomplishment during the reporting period was the development of a working fabrication method for silica/platinum Janus particles. The team established a procedure to create a uniform monolayer of silica microspheres on glass slides and then coat the exposed hemisphere of the particles with platinum. The use of a custom-built deposition system, supported by 3D printed components, allowed the team to control the motion, angle, and speed needed to form relatively uniform particle layers. This step was important because high-quality Janus particles are essential for reproducible assembly experiments. The particles were then characterized by microscopy, which confirmed successful formation of metallodielectric particles with one coated and one uncoated hemisphere. The team also developed and refined the experimental cell needed to observe Janus particle behavior under applied AC electric fields. Several design variations were tested, with the goal of reducing unwanted fluid movement and capillary effects that could interfere with particle assembly. The final cell design allowed small volumes of concentrated Janus particle suspension to be placed between glass slides and exposed to controlled electric fields while being observed under an optical microscope. This experimental platform provided a practical way to study how field strength and frequency influence the transition from dispersed particles to organized assemblies. Using this setup, the project generated an initial phase diagram for 6 wt% silica/platinum Janus particles suspended in water under AC electric fields. The team tested a range of electric field strengths and frequencies and identified four major particle-assembly regimes. At lower field strengths and lower frequencies, the particles remained mostly dispersed, indicating that the forces generated under those conditions were not strong enough to promote significant assembly. At intermediate conditions, the particles formed weak, loosely connected aggregates. At higher field strengths and selected frequencies, the particles assembled into chain-like structures. Under the strongest conditions tested, the particles formed denser bundled chains and began stacking into more three-dimensional structures. These assembly results are important because they demonstrate that the particle microstructure can be controlled by adjusting the applied electric field. Chain formation and dense aligned structures are especially relevant to the project's long-term goal of reinforcing 3D printed polymer materials. If similar structures can be produced and fixed inside UV-curable resins, they may provide a means of tailoring material properties in specific directions, such as strengthening a printed part along a load-bearing axis. The results also showed that silica/platinum Janus particles require different conditions than those reported for other particle systems, which reinforces the need to build a project-specific understanding of this material platform. In parallel with the experimental work, the team advanced the theoretical framework needed to interpret and guide the assembly process. The theoretical component focuses on modeling metallodielectric Janus particles and their interactions with external AC electric fields, as well as representing the surrounding polymer matrix. For the particle model, the team developed an approach that represents each Janus particle as a sphere with one conductive hemisphere and one dielectric hemisphere. The model is designed to account for how field strength, frequency, and material properties influence polarization and particle response. This provides a foundation for future comparisons between model predictions and observed particle assembly behavior.

The project also made progress on modeling the polymer environment, using a bead-spring representation of polymer chains. This type of model simplifies a polymer chain into connected segments while still capturing important physical behavior such as elasticity, relaxation, and chain configuration. The team validated the modeling approach by reproducing expected trends for ideal polymer chains and for real chains where self-avoidance is important. This progress is an important step toward connecting Janus particle assembly behavior with the behavior of the surrounding polymer resin, which will be necessary for predicting and optimizing reinforced printed materials.

Overall, the first-year work established the core experimental and computational capabilities needed for the next phase of the project. The team demonstrated successful particle fabrication, built an experimental system for observing field-directed assembly, identified multiple controllable assembly regimes, and developed initial theoretical models for both Janus particles and polymer chains. These accomplishments position the project to move from water-based assembly studies into UV-curable resin systems, where the particle structures can be incorporated into printable polymer composites.

During the next reporting period, the project will focus on translating the particle-assembly approach into resin-based materials. Planned work includes dispersing Janus particles in UV-curable resin, applying AC electric fields to organize the particles before curing, fabricating test specimens with different particle microstructures, and evaluating the mechanical performance of the resulting composites. The team will also study how the orientation of assembled particle chains relative to the applied load affects material properties. The theoretical work will continue in parallel, with emphasis on validating the particle model against prior work and beginning to assess possible polymer-particle blends. These next steps will directly support the project’s goal of enabling field-controlled reinforcement strategies for future 3D printing applications in space.

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